Sliding Puzzle Search
=====================

Enter the START board:
[[2, 4, 6, 8], [1, 3, 5, 7], [15, 14, 13, 11], [9, 10, 12, 0]]
Enter the GOAL board:
[[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [14, 13, 15, 0]]

Start and goal pass the solvability test.

1 = IDS
2 = Astar1 -- odd/even lookup table
3 = Astar2 -- misplaced tiles
4 = Astar3 -- Manhattan distance
5 = Astar4 -- max(h1,h3)
6 = Compare the four A* algorithms on random instances

Examples: 13, 2345, or 6
Algorithms to run: 45

==========
Astar3 -- Manhattan distance
==========
Number of moves : 30
Move sequence : LLULDRURUULDRDDLUURRULLLDRDRRD
Nodes removed from fringe : 2111
Maximum fringe size : 2001
CPU time : 0.026306 seconds

==========
Astar4 -- max(h1,h3)
==========
Number of moves : 30
Move sequence : LLULDRRUUULDRDLURRULLLDRDDRURD
Nodes removed from fringe : 1876
Maximum fringe size : 1841
CPU time : 39.180243 seconds

==========
RESTART: /Users/ravi/Desktop/cs480-fa24/code and test cases/astarNew.py
==========

Sliding Puzzle Search
=====================

Enter the START board:
[[2, 8, 4, 6], [1, 3, 5, 7], [15, 14, 13, 11], [9, 10, 12, 0]]
Enter the GOAL board:
[[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [14, 13, 15, 0]]

Start and goal pass the solvability test.

1 = IDS
2 = Astar1 -- odd/even lookup table
3 = Astar2 -- misplaced tiles
4 = Astar3 -- Manhattan distance

5 = Astar4 -- max(h1,h3)
6 = Compare the four A* algorithms on random instances

Examples: 13, 2345, or 6
Algorithms to run: 45

==========
Astar3 -- Manhattan distance
==========
Number of moves : 36
Move sequence : LLULDRRULDRURUULLDRULLDRDRURDLLURRDD
Nodes removed from fringe : 44753
Maximum fringe size : 40277
CPU time : 0.614868 seconds

==========
Astar4 -- max(h1,h3)
==========
Number of moves : 36
Move sequence : LLULDRRULDRURUULLDRULLDRDRURDLLURRDD
Nodes removed from fringe : 41973
Maximum fringe size : 38302
CPU time : 40.323999 seconds

==========
Sliding Puzzle Search
==========

Enter the START board:
[[5, 1, 2, 3], [9, 6, 11, 4], [13, 10, 7, 8], [14, 15, 12, 0]]
Enter the GOAL board:
[[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [14, 13, 15, 0]]

Start and goal pass the solvability test.

1 = IDS
2 = Astar1 -- odd/even lookup table
3 = Astar2 -- misplaced tiles
4 = Astar3 -- Manhattan distance
5 = Astar4 -- max(h1,h3)
6 = Compare the four A* algorithms on random instances

Examples: 13, 2345, or 6
Algorithms to run: 45

==========
Astar3 -- Manhattan distance
==========
Number of moves : 20
Move sequence : ULLLUURRRDDDLLURURDD
Nodes removed from fringe : 324
Maximum fringe size : 348
CPU time : 0.004697 seconds

==========
Astar4 -- max(h1,h3)

```
==========================================================================
Number of moves          : 20
Move sequence            : ULLLUURRRDDDLLURURDD
Nodes removed from fringe : 197
Maximum fringe size      : 217
CPU time                 : 39.928571 seconds

============================================================================================
RESTART: /Users/ravi/Desktop/cs480-fa24/code and test cases/astarNew.py
============================================================================================

Sliding Puzzle Search
=====================

Enter the START board:
[[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [13, 15, 14, 0]]
Enter the GOAL board:
[[0, 1, 2, 3], [4, 5, 6, 7], [8, 9, 10, 11], [12, 13, 14, 15]]

Start and goal pass the solvability test.

1 = IDS
2 = Astar1 -- odd/even lookup table
3 = Astar2 -- misplaced tiles
4 = Astar3 -- Manhattan distance
5 = Astar4 -- max(h1,h3)
6 = Compare the four A* algorithms on random instances

Examples: 13, 2345, or 6
Algorithms to run: 45

==========================================================================
Astar3 -- Manhattan distance
==========================================================================
Number of moves          : 44
Move sequence            : LURULLDRRUULDRDLULLDRRRDLLLURRRDLUUULDLDRUUL
Nodes removed from fringe : 2910404
Maximum fringe size      : 2371147
CPU time                 : 145.203205 seconds

==========================================================================
Astar4 -- max(h1,h3)
==========================================================================
Number of moves          : 44
Move sequence            : LURULLDRRUULDRDDLUULLDRRDLLURRRDLUULLDRURULL
Nodes removed from fringe : 1310162
Maximum fringe size      : 1181025
CPU time                 : 98.192376 seconds
```

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Sliding Puzzle Search
=====================

Enter the START board:

================================================================================ RESTART: /Users/ravi/Desktop/cs480-

Sliding Puzzle Search
=====================

Enter the START board:
[[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [13, 15, 14, 0]]
Enter the GOAL board:
[[4, 8, 12, 0], [3, 7, 11, 15], [2, 6, 10, 14], [1, 5, 9, 13]]

Start and goal pass the solvability test.

1 = IDS
2 = Astar1 -- odd/even lookup table
3 = Astar2 -- misplaced tiles
4 = Astar3 -- Manhattan distance
5 = Astar4 -- max(h1,h3)
6 = Compare the four A* algorithms on random instances

Examples: 13, 2345, or 6
Algorithms to run: 45

=================================================================
Astar3 -- Manhattan distance
=================================================================
Number of moves           : 39
Move sequence             : LLLUUURRRDDLDLLUUURRRDDLLURRDDLLLUUURRR
Nodes removed from fringe : 735
Maximum fringe size       : 615
CPU time                  : 0.008399 seconds

=================================================================
Astar4 -- max(h1,h3)
=================================================================
Number of moves           : 39
Move sequence             : LLLUUURRRDDLDLLUUURRRDDLLURRDDLLLUUURRR
Nodes removed from fringe : 719
Maximum fringe size       : 606
CPU time                  : 59.703788 seconds
>>> Tab = oddEven([[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [13, 15, 14, 0]])
...
>>> Tab
...
```